SANTHEEPKUMAR VEERASAMY

PRINCIPAL SOFTWARE ENGINEER | ENTERPRISE SYSTEMS ARCHITECT & TECHNICAL LEADER

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OVERVIEW

I'm a self-taught, obsessive software builder with 7+ years of experience taking complex systems from a blank repository to production cloud infrastructure. My journey started in a government school in Tamil Nadu on a 2GB RAM state-issued laptop, teaching myself programming through documentation and YouTube—a drive that took me from converting a college internship into a full-time role in under 3 months (Skematix), to fast-tracking from Senior Engineer to Principal Software Engineer (HOD Digitals), leading tier-1 enterprise deliveries at Heady (shipping a Chuck E. Cheese web MVP solo in 2 months vs. an 8-month estimate), and now owning end-to-end architecture as a Staff Engineer at Tuna Software Solution and Lead Software Engineer (Part-Time Contractor) at Science & Humans.

As a Principal-level architect and hands-on builder, I own the entire technical lifecycle: designing distributed microservices & inter-service communication layers, engineering authentication/RBAC and multi-channel notification engines from the ground up, orchestrating Kubernetes (K8s) and AWS infrastructure with automated CI/CD, and mentoring engineering teams to ship with high velocity (+35%). Outside of client work, I constantly build and ship my own production platforms—including StrawHatDevs (strawhatdevs.com + 200+ architecture and programming videos on YouTube), ChimeFlow (a multi-tenant omnichannel notification & cadence platform), ENVault (a zero-knowledge encrypted .env sync platform with a native Go CLI), and IncidentLens AI (an evidence-grounded incident triage & Runbook RAG engine).

CORE TECHNICAL COMPETENCIES

Cloud, K8s & DevOps:	Kubernetes (K8s), Docker, AWS (ECS, Lambda, S3, RDS, CloudFront, IAM), GCP, GitHub Actions, Bitbucket Pipelines, Jenkins, Vercel, Terraform / IaC, Sentry, Axiom
Architecture & Systems:	Microservices, Inter-Service Communication, Event-Driven Systems, Clean/Hexagonal Architecture, Monorepos (Turborepo), Auth/RBAC, Notification Engines, Redis, Kafka, RabbitMQ
Frontend Engineering:	React, Next.js (App & Pages Router), TypeScript, JavaScript (ES6+), SvelteKit / Svelte 5, React Native, Expo, Tailwind CSS, State Management (Zustand, Redux)
Backend & Languages:	Node.js, NestJS, Go (Golang), Bun, Express.js, RESTful APIs, GraphQL, WebCrypto / AES-256-GCM Encryption, IoT Hardware & POS Peripherals
Databases & Storage:	PostgreSQL (Raw SQL, RLS, FTS/tsvector, SKIP LOCKED), MongoDB, Supabase, Drizzle ORM, TypeORM, Prisma, Sequelize, Mongoose, Query Tuning
Testing, AI & Quality:	Playwright, Vitest, Jest, React Testing Library, LLM RAG & Bounded Agents, Automated Eval Harnesses, CI/CD Quality Gates

PROFESSIONAL EXPERIENCE

Staff Engineer | Tuna Software Solution
Jun 2025 – Present

- **End-to-End Architecture & Delivery:** Owned full-stack architecture, cloud infrastructure, backend microservices, and frontend platforms across two concurrent enterprise initiatives, driving feature releases on schedule.
- **Pony Up IoT Kiosk & Cloud Web Migration:** Architected and delivered an Android self-service kiosk (React Native/Expo) integrating physical IoT peripherals (payment terminals, receipt printers, cash drawers), then led its rebuild as a cloud web platform to eliminate deployment friction and improve cross-device compatibility.
- **CI/CD Automation & Engineering Standards:** Established structured code review processes, automated CI/CD pipelines (GitHub Actions/Docker), and coding standards—boosting team delivery velocity by 35% and reducing production defect escapes.
- **Stakeholder & Product Alignment:** Collaborated directly with product stakeholders to translate business requirements into scalable technical solutions, acting as the primary technical point of contact.

Lead Software Engineer (Part-Time Contractor) | Science & Humans
Jun 2025 – Present

- **Ground-Up Ownership & Product Planning:** Contributed from the ground up with full end-to-end ownership—partnering directly with Product Managers to plan implementations, estimate modules, architect solutions, and ship production releases.
- **Authentication, Authorization & Notification Engine:** Architected and built the entire authentication and role-based authorization (RBAC) system alongside a centralized multi-channel notification system for the platform.
- **Microservices & Communication Layer:** Set up decoupled backend microservices and engineered the inter-service communication layer from scratch while resolving critical system bottlenecks and bugs to harden platform stability.
- **Kubernetes (K8s) & DevOps Configuration:** Configured and managed containerized application deployments on Kubernetes (K8s) and built automated CI/CD DevOps workflows for reliable, zero-downtime releases.
- **Mentoring & Team Ramp-Up:** Guided, onboarded, and mentored junior developers joining the project, conducting code reviews and unblocking engineering execution across modules.

Senior Software Engineer | Heady
May 2023 – Jun 2025

- **Enterprise Client Leadership:** Served as primary technical lead interfacing with tier-1 enterprise clients (HeyDay Skin Care, Chuck E. Cheese, The Lash Lounge), coordinating with PMs and stakeholders to define scope, set milestones, and ship production systems.
- **Chuck E. Cheese Web MVP (75% Faster Delivery):** Delivered the Chuck E. Cheese web kiosk MVP in 2 months as a solo contributor, compressing an 8-month original project timeline by 75% through focused full-stack engineering and rapid iteration.
- **HeyDay Skin Care Microservices Ecosystem:** Architected a microservices-based backend integration with Boulevard (booking platform), Redis caching, Sentry error monitoring, and AWS infrastructure sustaining 99.9% reliability.
- **AI-Assisted Workflows & Mentorship:** Mentored 6+ junior and mid-level developers on modern development practices, AI-assisted coding workflows, and architectural patterns that improved code quality and reduced regression rates.

Principal Software Engineer (Promoted from Senior Frontend Engineer) | HOD Digitals
Sep 2021 – May 2023

- **Fast-Track Promotion (Senior Engineer → Principal Engineer):** Joined as Senior Frontend Engineer contributing to Project Gen and concurrent client systems; promoted to Principal Software Engineer (Nov 2022) to lead technical direction and architecture across multiple product lines.
- **E-Commerce & Scalable System Architecture:** Architected and shipped full-stack e-commerce platforms (Tamil Mart web & iOS/Android mobile apps with Next.js, Magento GraphQL, React Native), CRM subscription systems (Vmeals on AWS), and Playwright CI/CD test automation (AMS).

- **Cross-Functional Leadership & Mentorship:** Oversaw project planning, system design, and cloud infrastructure decisions while acting as primary technical liaison between engineering teams and business stakeholders.

Full Stack Developer (Converted from Intern in <3 Months) | Skematix Technologies — Dec 2019 – Sep 2021

- **Rapid Intern-to-Full-Time Conversion:** Joined during college in 2019 as an Intern Software Engineer and earned conversion to full-time Full Stack Developer within 2–3 months due to rapid execution and deep ownership.
- **End-to-End Web Application Delivery:** Developed, optimized, and deployed multiple client-facing web applications across React, Node.js, Express, and relational/NoSQL databases from initial schema design to production hosting.

SELF-BUILT PRODUCTS, PLATFORMS & DEVELOPER COMMUNITY

ChimeFlow — Multi-Tenant Omnichannel Notification & Cadence Platform

Engineered a multi-tenant notification & cadence orchestration platform supporting Email (SendGrid), SMS (Twilio), and Push (Firebase FCM) using Clean/Hexagonal Architecture. Built a write-ahead delivery ledger, fair-claim background worker queues, idempotency keys, per-tenant rate limiting, suppression/consent engines, encrypted credentials at rest, and Razorpay billing.

Bun, TypeScript, SvelteKit / Svelte 5, PostgreSQL 17, Clerk Auth, SendGrid, Twilio, Firebase Admin, Docker, AWS

ENVault — Zero-Knowledge Encrypted .env Sync Platform & Native Go CLI

Built an end-to-end client-encrypted team environment variable & secret synchronization platform with browser WebCrypto key derivation, atomic revision rollback, and a native cross-platform Go CLI (github.com/Santheepkumar/envault-cli) integrated with macOS Keychain, Windows Credential Manager, and Linux Secret Service.

Go (Golang CLI), Next.js, TypeScript, PostgreSQL 17, Drizzle ORM, WebCrypto API, Better Auth, Docker

IncidentLens AI — Evidence-Grounded Incident Triage & Runbook RAG Engine

Designed a multi-tenant AI production incident investigator connecting Sentry telemetry with Markdown runbooks (PostgreSQL tsvector + GIN Hybrid FTS RAG), AES-256-GCM secret vault, a bounded 6-step agent loop with deterministic [ev_xxx] citation verification, FOR UPDATE SKIP LOCKED job queues, and a 35-case AI evaluation harness (93.3% root-cause accuracy).

TypeScript, Turborepo, Node.js, PostgreSQL (Raw SQL + RLS), Auth0 OAuth2 PKCE, LLM Tool-Calling & RAG, Vitest

StrawHatDevs Platform (strawhatdevs.com) & Route-Atlas Open-Source Tooling

Architected strawhatdevs.com full-stack learning platform (200+ architecture/programming videos) alongside developer tools including Route-Atlas (published npm visual atlas for Next.js App Router) and SelectCheck (Tauri v2 + Cloudflare Workers AI assistant).

Next.js 15, React 19, TypeScript, Supabase, PostgreSQL, Sanity CMS, Tauri v2, Cloudflare Workers, Vercel CI/CD

KEY ENTERPRISE PROJECTS & PRODUCTION DEPLOYMENTS

Science & Humans — Healthcare Platform & Microservices Ecosystem

Built core platform architecture from the ground up with full end-to-end ownership: engineered the complete authentication & RBAC authorization system, event-driven notification engine, decoupled microservices & inter-service communication layer, and production Kubernetes (K8s) + CI/CD DevOps infrastructure while mentoring junior developers.

TypeScript, Node.js, Microservices, Auth/RBAC, Notification System, Kubernetes (K8s), Docker, CI/CD

Pony Up — Android Self-Service IoT Retail Kiosk & Cloud Web Platform

Engineered a React Native/Expo self-service kiosk deployed in shopping malls integrating physical IoT hardware (payment terminals, thermal receipt printers, cash drawers), and subsequently led its strategic rebuild into a cloud-hosted web platform.

React Native, Expo, Next.js, IoT Peripherals, Android Kiosk Mode, REST APIs, Docker, GitHub Actions

Chuck E. Cheese — Customer Experience Web Platform (2-Month Solo MVP)

Rebuilt the Chuck E. Cheese kiosk into a mobile-responsive customer experience web platform enabling instant product interaction via QR code scanning—delivered the production MVP solo in 2 months vs. an 8-month original estimate (75% faster time-to-market).

React, Next.js, TypeScript, Tailwind CSS, QR Integration, AWS CloudFront, Automated CI/CD

HeyDay Skin Care & The Lash Lounge — Enterprise Booking & Microservices

Architected high-concurrency appointment booking widgets, e-commerce shop apps, and microservices backends integrated with Boulevard API, Redis caching, Sentry monitoring, and AWS infrastructure sustaining 99.9% uptime.

React, Next.js, Node.js, NestJS, Redis, Microservices, AWS, Vercel, Bitbucket Pipelines, Sentry, Boulevard API

Additional Production Platforms & Enterprise Deliveries (10+ End-to-End Systems Shipped)

Beyond the flagship systems above, architected, led, and shipped numerous production platforms across Healthcare, B2B SaaS, E-Commerce, CRM, and CI/CD Automation—including Mobilab2u (healthcare appointment & bloodwork scheduling microservices platform on AWS ECS), Nest NCER (entrepreneur sustainable management B2B SaaS with real-time analytics dashboards), Project Gen (enterprise digital platform), The Lash Lounge (national beauty chain booking platform), Vmeals (automated CRM & subscription customization platform), Tamil Mart (omnichannel e-commerce web & iOS/Android mobile app), AMS (CI/CD release & Playwright test automation platform), HOD Ecom PWA (reusable Next.js + Magento storefront framework replacing ScandiPWA), Haladoc (B2B doctor appointment web & mobile platform), and Medisouq (pharmaceutical e-commerce marketplace).

NestJS, Next.js, React, React Native, Expo, Node.js, Express, PostgreSQL, MongoDB, Redis, Magento GraphQL, Apollo, ScandiPWA, Playwright, Docker, AWS ECS, Vercel, GitHub Actions, Bitbucket Pipelines

EDUCATION

Bachelor of Engineering (B.E.) — Computer Science & Engineering • Anna University, Tamil Nadu, India — 2015 – 2019